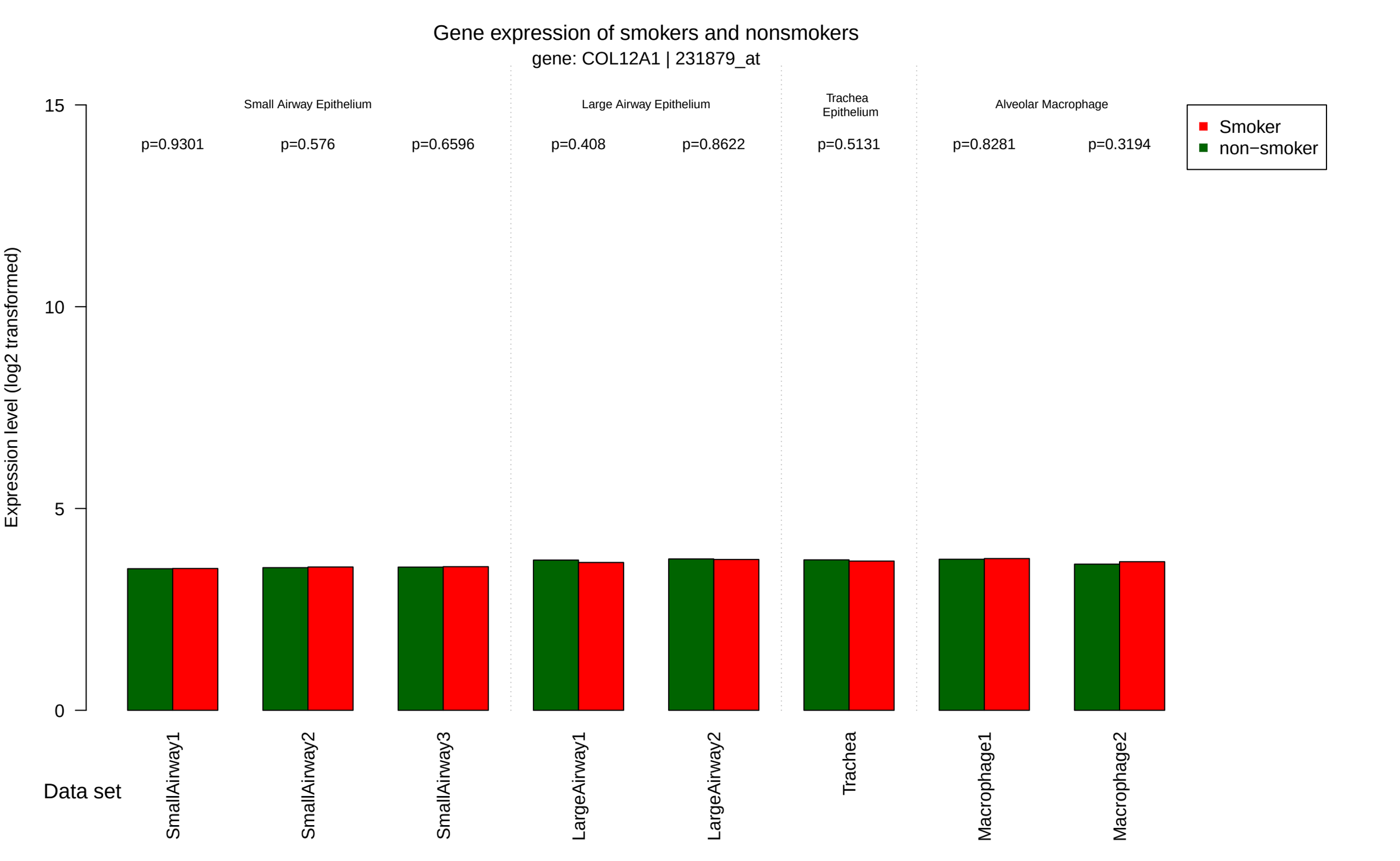
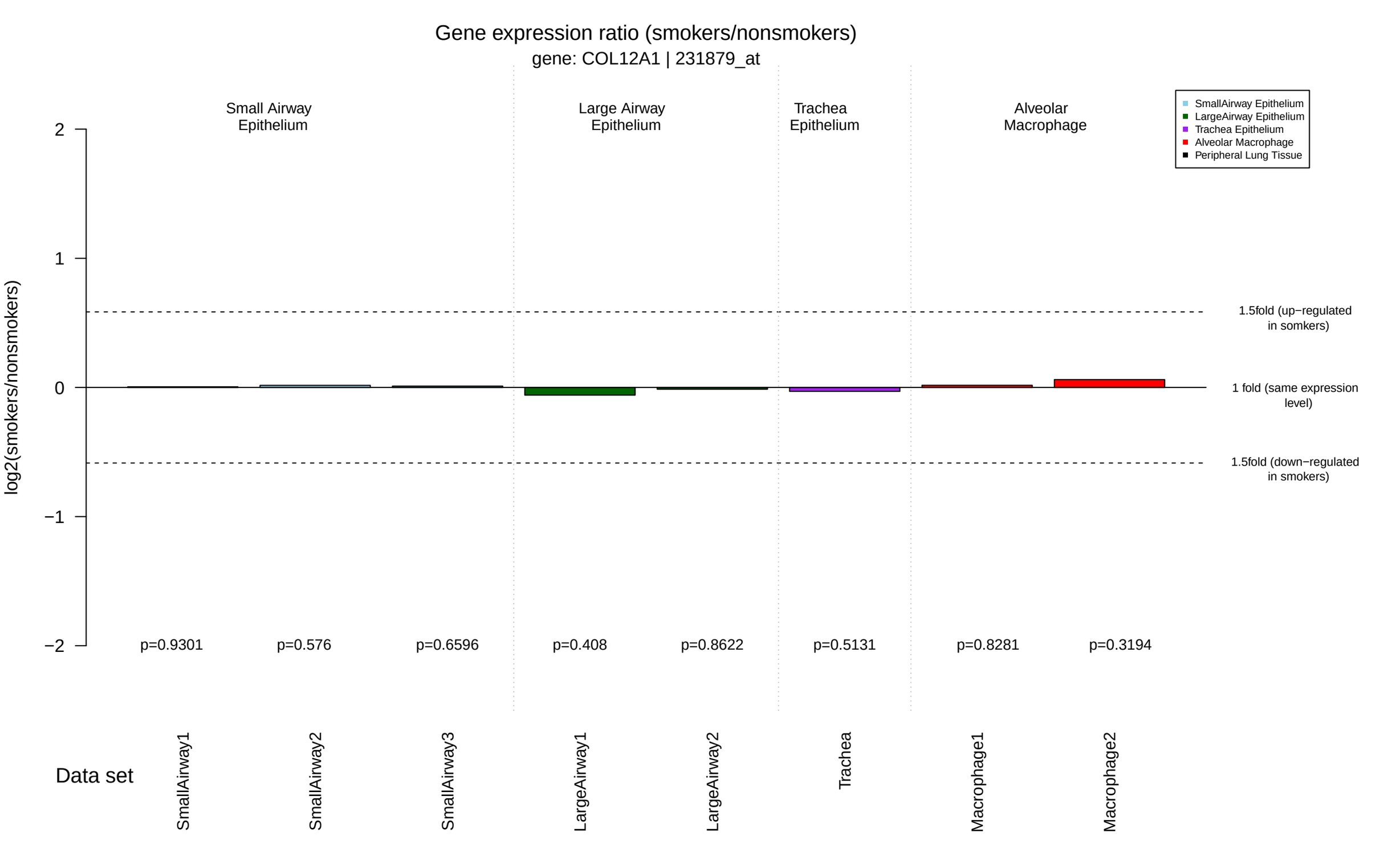
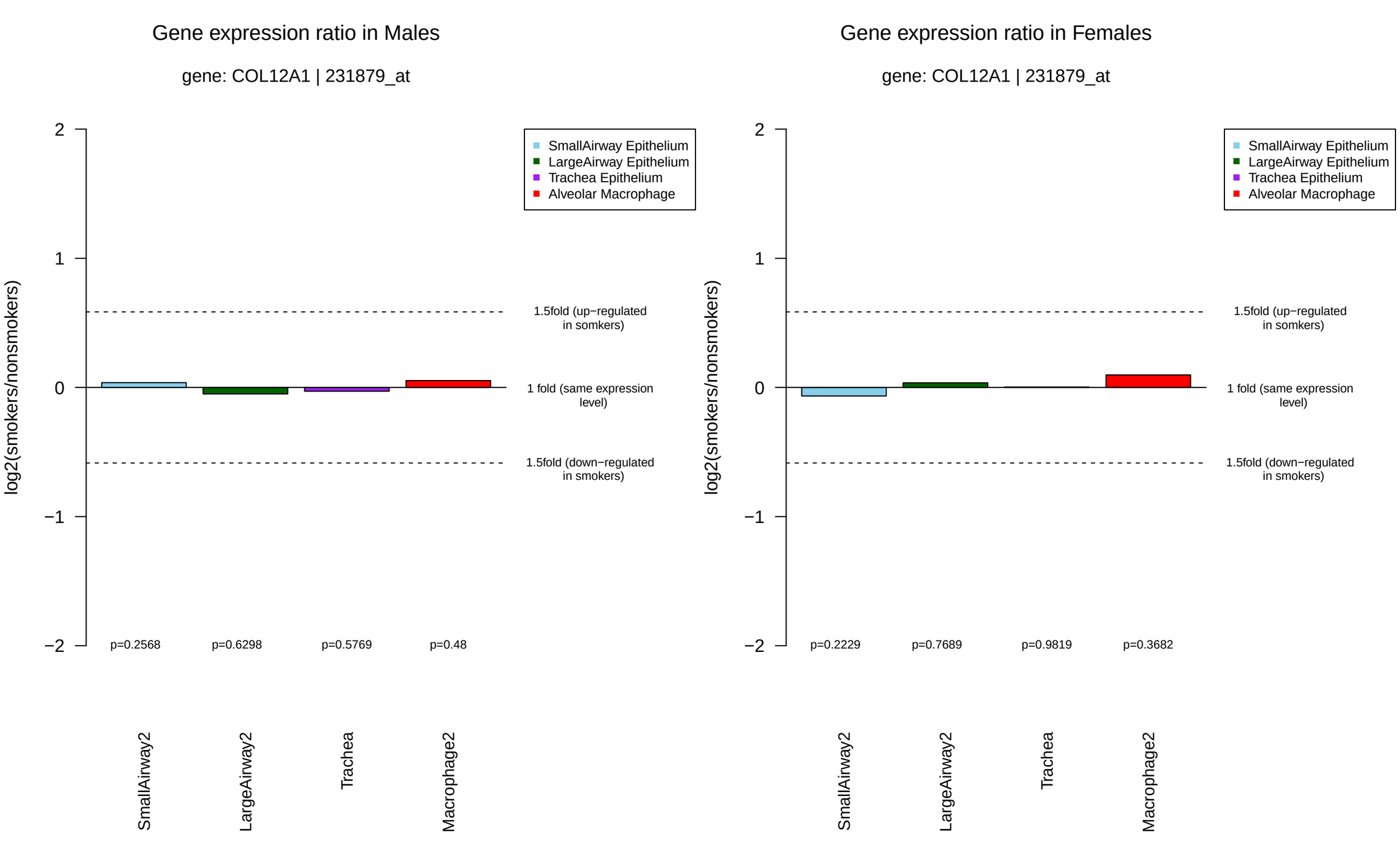
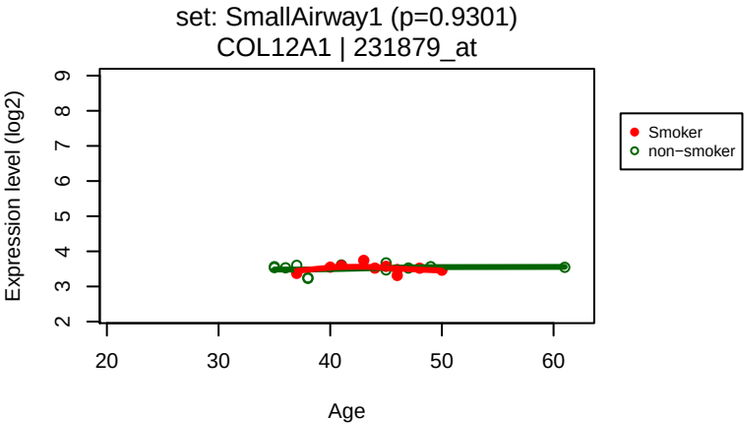
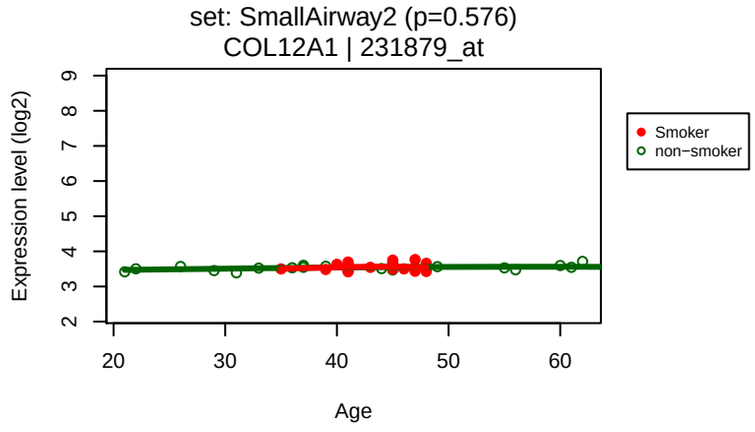
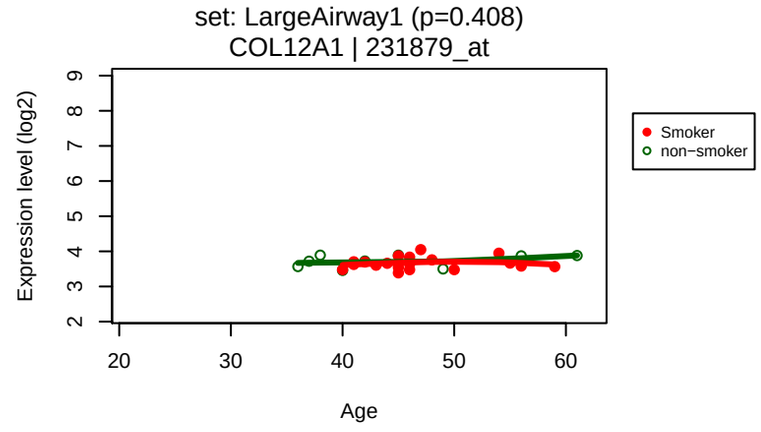

Gene expression of smokers and nonsmokers
gene: COL12A1 | 231879_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9301
p=0.576
p=0.6596
p=0.408
p=0.8622
p=0.5131
p=0.8281
p=0.3194
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

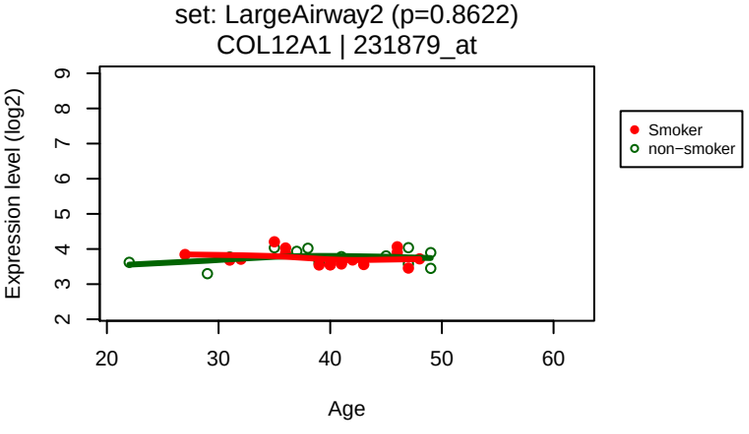

Gene expression ratio (smokers/nonsmokers)
gene: COL12A1 | 231879_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9301
p=0.576
p=0.6596
p=0.408
p=0.8622
p=0.5131
p=0.8281
p=0.3194
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

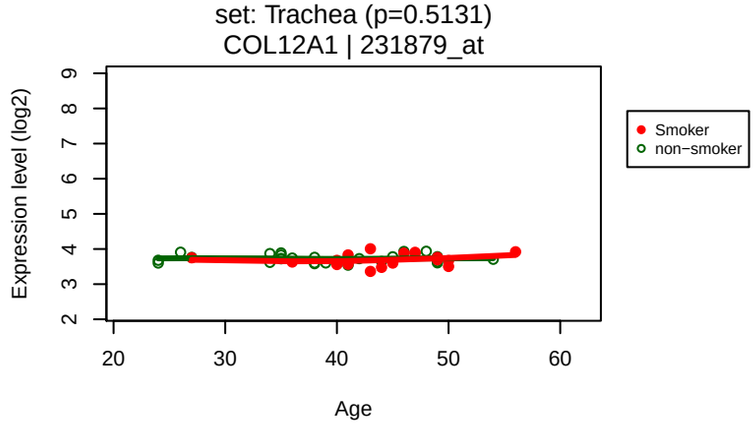

Gene expression ratio in Males
gene: COL12A1 | 231879_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2568
p=0.6298
p=0.5769
p=0.48
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL12A1 | 231879_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2229
p=0.7689
p=0.9819
p=0.3682
SmallAirway2
LargeAirway2
Trachea
Macrophage2

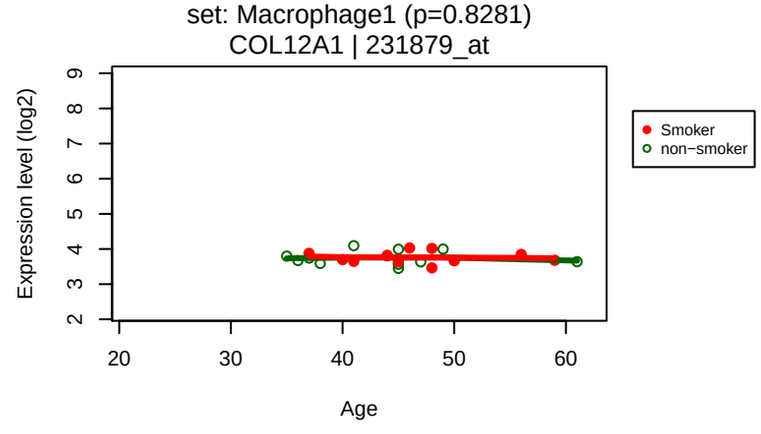

set: SmallAirway1 (p=0.9301)
COL12A1 | 231879_at
Expression level (log2)
Age
Smoker
non−smoker

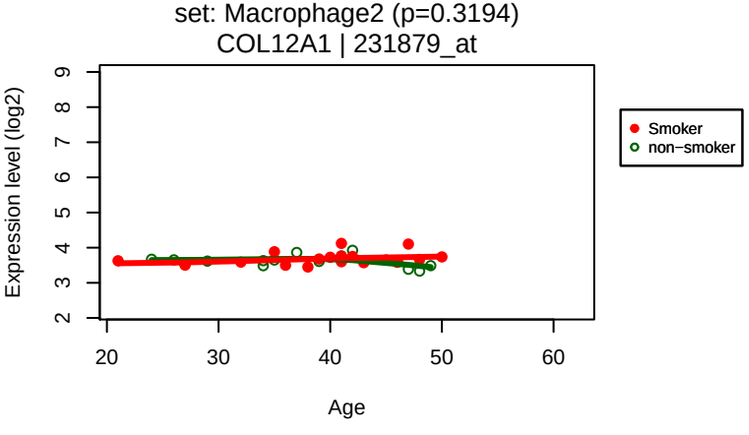

set: SmallAirway2 (p=0.576)
COL12A1 | 231879_at
Expression level (log2)
Age
Smoker
non−smoker

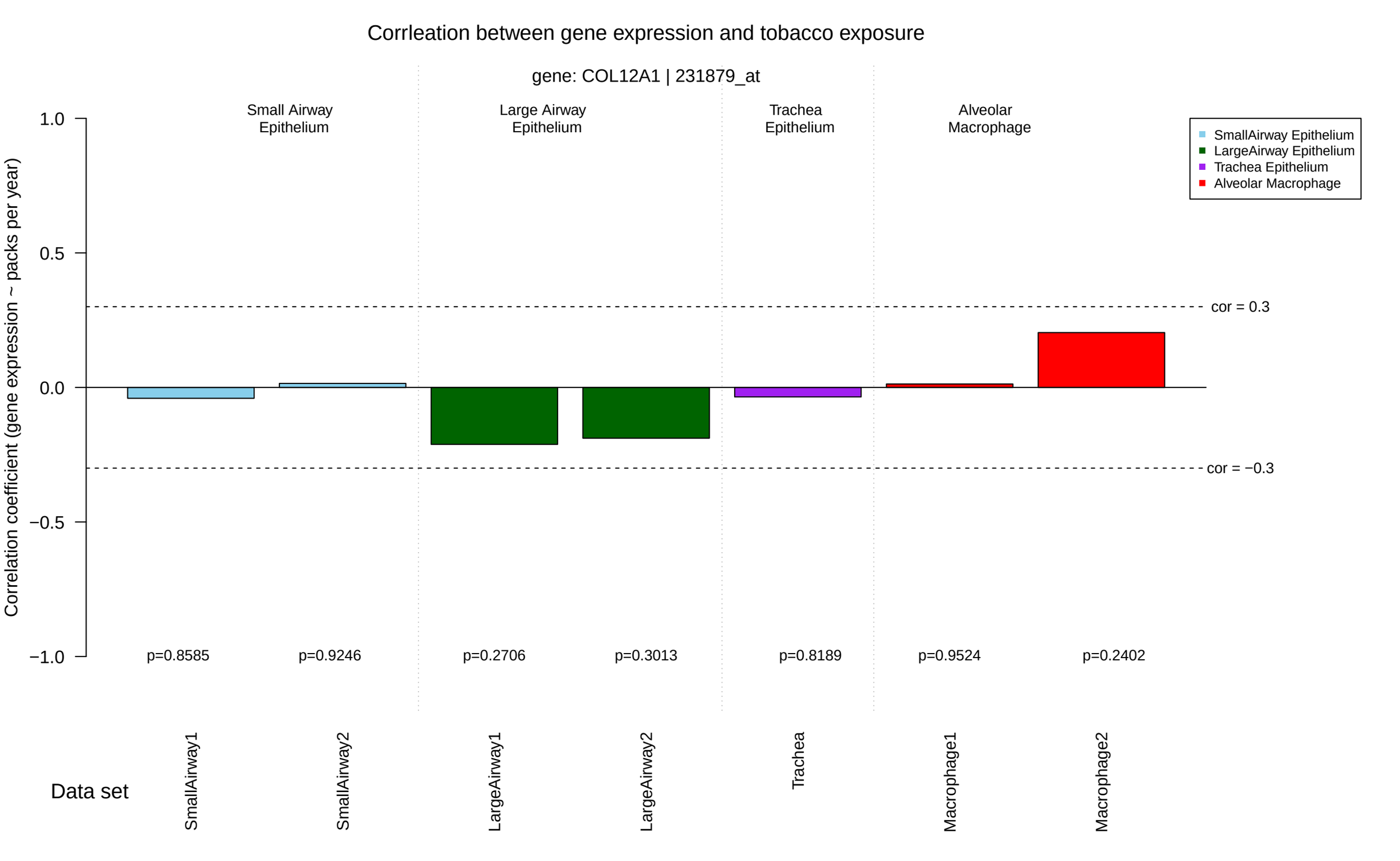

set: LargeAirway1 (p=0.408)
COL12A1 | 231879_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8622)
COL12A1 | 231879_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5131)
COL12A1 | 231879_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8281)
COL12A1 | 231879_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3194)
COL12A1 | 231879_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL12A1 | 231879_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8585
p=0.9246
p=0.2706
p=0.3013
p=0.8189
p=0.9524
p=0.2402
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2